

# Gap-Weighted Weak-Mode Quotients

## Adaptive Ritz Enrichment Without Unstable Direction Recovery

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### Abstract

RH-95 showed that the fourth projected-cross direction can be almost null: its singular value may be eleven orders of magnitude below the leading mode. Geometric reconstruction is then unstable, even though the corrected Ritz tail is insensitive to the direction. We replace weak-direction identification by an energy quotient.

For a positive block compression

$$H = \begin{pmatrix} A & C \\ C^* & D \end{pmatrix},$$

let  $\Phi_r(H)$  denote the sum of its largest  $r$  eigenvalues. We first prove the universal omitted-block bound

$$0 \leq \Phi_r(H) - \Phi_r(A) \leq 2 \|C\|_* + \text{tr } D.$$

More sharply, suppose the retained rank- $r$  cutoff satisfies  $\lambda_r^\downarrow(A) \geq \alpha$ , while  $D \leq \beta I$  with  $\alpha > \beta$ . Our gap-weighted weak-mode tail-loss theorem gives

$$0 \leq \Phi_r(H) - \Phi_r(A) \leq \frac{\|C\|_F^2}{\alpha - \beta}.$$

The proof optimizes the amount of a rank- $r$  projector entering the omitted block and requires no identification of its weak basis vectors.

We audit adaptive projected-cross widths at relative singular cutoffs  $10^{-8}, 10^{-6}, 10^{-4}$  over the 120 source-seeded updates. At the primary cutoff  $10^{-8}$ , exactly five weak fourth modes are omitted. All five losses are gap-certified, all updates remain Ritz-monotone, and all ten endpoints stay below the 1.01 target; the worst endpoint/reference ratio is 1.001173. The largest adaptive/full one-step tail ratio is 1.00000465. More aggressive cutoffs reveal the next obstruction. Every local omission is still certified, but the recursive endpoint target fails: the worst ratios are 1.024922 at  $10^{-6}$  and 1.014092 at  $10^{-4}$ .

Thus the ill-conditioned fourth mode can be quotiented safely at the five identified updates, but local certificates do not compose automatically. The next task is a horizon recurrence for accumulated quotient losses. No uniform gap theorem, Hilbert–Polya operator, zero identification, or Riemann Hypothesis result is claimed.

**Keywords:** Ky Fan sum; Ritz enrichment; weak singular mode; spectral gap; adaptive rank; validated numerics.

**MSC 2020:** 47A75; 15A18; 65F15; 65G20; 37C30.

## 1 Introduction

The source-seeded chain of RH-94 uses four projected-cross directions to track the ambient leading packet over a complete finite prefix [5]. RH-95 reduced the ambient cross SVD to packet-scale normal matrices, but found that the fourth cross mode can be extremely ill conditioned [4]. In the weakest updates, different stabilized fourth directions produce visibly different projectors but nearly identical corrected tails.

This suggests that geometric identification is stronger than the route needs. The packet is used to control captured energy and subsequent tail propagation. If adjoining a weak direction can improve the leading rank- $r$  capture by at most a certified tolerance, then that direction should be quotiented rather than reconstructed.

The basic finite-dimensional problem is as follows. A retained enrichment produces a positive matrix  $A$ . Additional weak directions enlarge it to

$$H = \begin{pmatrix} A & C \\ C^* & D \end{pmatrix}.$$

How much can the leading rank- $r$  Ky Fan sum increase? A universal answer comes from trace-norm perturbation. A sharper answer uses two pieces of structure: the  $r$ th retained Ritz value lies above the omitted diagonal spectrum, and the coupling  $C$  is small.

The resulting gap-weighted estimate is elementary but useful. It bounds the energy value of the entire omitted subspace without selecting a stable basis inside that subspace. It therefore matches the exact obstruction exposed by RH-95.

The audit then asks a second question: how aggressively can weak modes be removed along the recursive source-to-endpoint chain? A cutoff at  $10^{-8}$  removes exactly the five directions already identified as unstable and preserves every endpoint. Larger cutoffs fail, even though each individual loss remains locally certified. This separates local quotient validity from global horizon composition.

## 2 Ky Fan formulation

For a Hermitian matrix  $X$ , define

$$\Phi_r(X) = \sum_{j=1}^r \lambda_j^\downarrow(X) = \max_{\substack{P=P^*=P^2 \\ \text{tr } P=r}} \text{tr}(PX).$$

The variational equality is Ky Fan's maximum principle [1, 2].

Let

$$H = \begin{pmatrix} A & C \\ C^* & D \end{pmatrix} \geq 0, \quad A \in \mathbb{C}^{m \times m}, \quad m \geq r. \quad (1)$$

The retained Ritz construction uses  $\Phi_r(A)$ , whereas adding the weak directions makes  $\Phi_r(H)$  available. Their difference is exactly the captured-energy loss caused by quotienting the omitted block.

**Theorem 2.1** (Universal omitted-block bound). *For (1),*

$$0 \leq \Phi_r(H) - \Phi_r(A) \leq 2 \|C\|_* + \text{tr } D. \quad (2)$$

*Proof.* Embed  $A$  as  $H_0 = \text{diag}(A, 0)$ . Since  $A \geq 0$  and  $m \geq r$ ,  $\Phi_r(H_0) = \Phi_r(A)$ . Enlargement of the trial space gives the lower bound. The Ky Fan sum is Lipschitz in trace norm:

$$|\Phi_r(H) - \Phi_r(H_0)| \leq \|H - H_0\|_*.$$

Now

$$H - H_0 = \begin{pmatrix} 0 & C \\ C^* & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & D \end{pmatrix}.$$

The first matrix has trace norm  $2 \|C\|_*$ , and the second has trace norm  $\text{tr } D$  because  $D \geq 0$ .  $\square$

The universal bound needs no gap, but it counts all omitted diagonal energy. For a weak direction whose diagonal energy is not itself small, it can be too large for a relative tail budget.

### 3 Gap-weighted weak-mode tail-loss theorem

**Theorem 3.1** (Gap-weighted weak-mode tail-loss theorem). *Assume in (1) that*

$$\lambda_r^\downarrow(A) \geq \alpha, \quad D \leq \beta I, \quad \delta := \alpha - \beta > 0.$$

*Then*

$$0 \leq \Phi_r(H) - \Phi_r(A) \leq \frac{\|C\|_F^2}{\delta}. \quad (3)$$

*Proof.* Let  $P$  be any rank- $r$  orthogonal projector on the full block space and write

$$P = \begin{pmatrix} P_{11} & P_{12} \\ P_{12}^* & P_{22} \end{pmatrix}, \quad t = \operatorname{tr} P_{22}.$$

Since  $0 \leq P_{11} \leq I$  and  $\operatorname{tr} P_{11} = r - t$ , the variational description of partial spectral mass gives

$$\operatorname{tr}(P_{11}A) \leq \Phi_r(A) - \alpha t. \quad (4)$$

Also  $\operatorname{tr}(P_{22}D) \leq \beta t$ . From  $P^2 = P$ ,

$$P_{12}^* P_{12} = P_{22} - P_{22}^2,$$

so  $\|P_{12}\|_F^2 \leq t$ . Therefore

$$\begin{aligned} \operatorname{tr}(PH) - \Phi_r(A) &\leq -\delta t + 2 \operatorname{Re} \operatorname{tr}(P_{12}C^*) \\ &\leq -\delta t + 2 \|C\|_F \sqrt{t} \\ &\leq \frac{\|C\|_F^2}{\delta}. \end{aligned}$$

The last line maximizes the scalar quadratic in  $\sqrt{t}$ . Taking the maximum over rank- $r$  projectors  $P$  proves the upper bound. The lower bound again follows by retaining the old trial space.  $\square$

*Remark 3.2.* The theorem identifies the correct quotient data. It does not require the omitted cross singular value itself, nor a stable orientation of its left singular vector. It requires only a coupling norm and a retained-to-omitted compressed spectral gap.

#### 3.1 Adaptive width

Let

$$s_1(K) \geq s_2(K) \geq \dots$$

be projected-cross singular values. Given a relative threshold  $\tau$  and width limits  $k_{\min} \leq k_{\max}$ , define

$$k_\tau = \max \{k_{\min}, \min(k_{\max}, \#\{j \leq k_{\max} : s_j(K)/s_1(K) \geq \tau\})\}. \quad (5)$$

The retained enrichment uses the first  $k_\tau$  directions. The remaining directions up to  $k_{\max}$  form the omitted block in Theorem 3.1.

The rule is deliberately local. It says which directions are weak at one update. It does not yet allocate a global horizon error budget.

Table 1: Adaptive threshold summary over 120 updates.

| $\tau$    | width 2 | width 3 | width 4 | omitted | worst endpoint |
|-----------|---------|---------|---------|---------|----------------|
| $10^{-8}$ | 0       | 5       | 115     | 5       | 1.001173       |
| $10^{-6}$ | 1       | 10      | 109     | 11      | 1.024922       |
| $10^{-4}$ | 5       | 17      | 98      | 22      | 1.014092       |

## 4 Validated audit

### 4.1 Protocol

We replay the five scales and two directional channels of RH-94, from the source seed to endpoints 4, 6, 11, 17, 22. The rank clock is unchanged, with minimum and maximum enrichment widths two and four. Three relative cutoffs are tested:

$$\tau \in \{10^{-8}, 10^{-6}, 10^{-4}\}.$$

Each threshold runs its own recursive chain, so endpoint behavior need not be monotone in  $\tau$ .

At every omitted update, the retained and full-width bases are assembled in one ordered QR factorization. The compressed blocks  $A, C, D$  are then extracted from the same full matrix. Binary64 eigendecompositions are inflated by their reconstruction residuals and a roundoff guard to obtain a lower bound for  $\alpha$  and an upper bound for  $\beta$ . Tail forms are evaluated from exact binary inputs in Arb at 384-bit precision [3].

### 4.2 Primary cutoff

At  $\tau = 10^{-8}$ , the adaptive chain omits the fourth direction in exactly five of 120 updates. These are the five weak-mode events identified by the conditioning audit of RH-95. Every omitted update has a positive certified gap and satisfies (3).

The chain uses width three five times and width four 115 times. All direct Ritz steps are monotone. The largest adaptive/full-width one-step tail ratio is

$$1.0000046481.$$

All ten endpoint/reference ratios remain below 1.01, with maximum

$$1.0011723197.$$

Thus unstable fourth-mode inversion is unnecessary at precisely the updates where it is least meaningful.

The gap bounds are conservative. The largest primary bound is about 5085 times its observed one-step loss, but remains below  $2.90 \times 10^{-4}$  of the corresponding full-width tail. This is adequate for the finite local decision.

### 4.3 Aggressive cutoffs and the composition barrier

The two larger thresholds fail the endpoint gate, as shown in Table 1. This does not come from an invalid local theorem: all eleven omissions at  $10^{-6}$  and all twenty-two omissions at  $10^{-4}$  have green gap certificates. Instead, small one-step losses alter the packet fed into later updates and accumulate nonlinearly.

The recursive response is not monotone in the threshold. The  $10^{-6}$  chain has a worse endpoint than the  $10^{-4}$  chain even though it omits fewer directions. Once the packet paths diverge, a larger local trial space at one time does not imply a nested trial space at all later times.

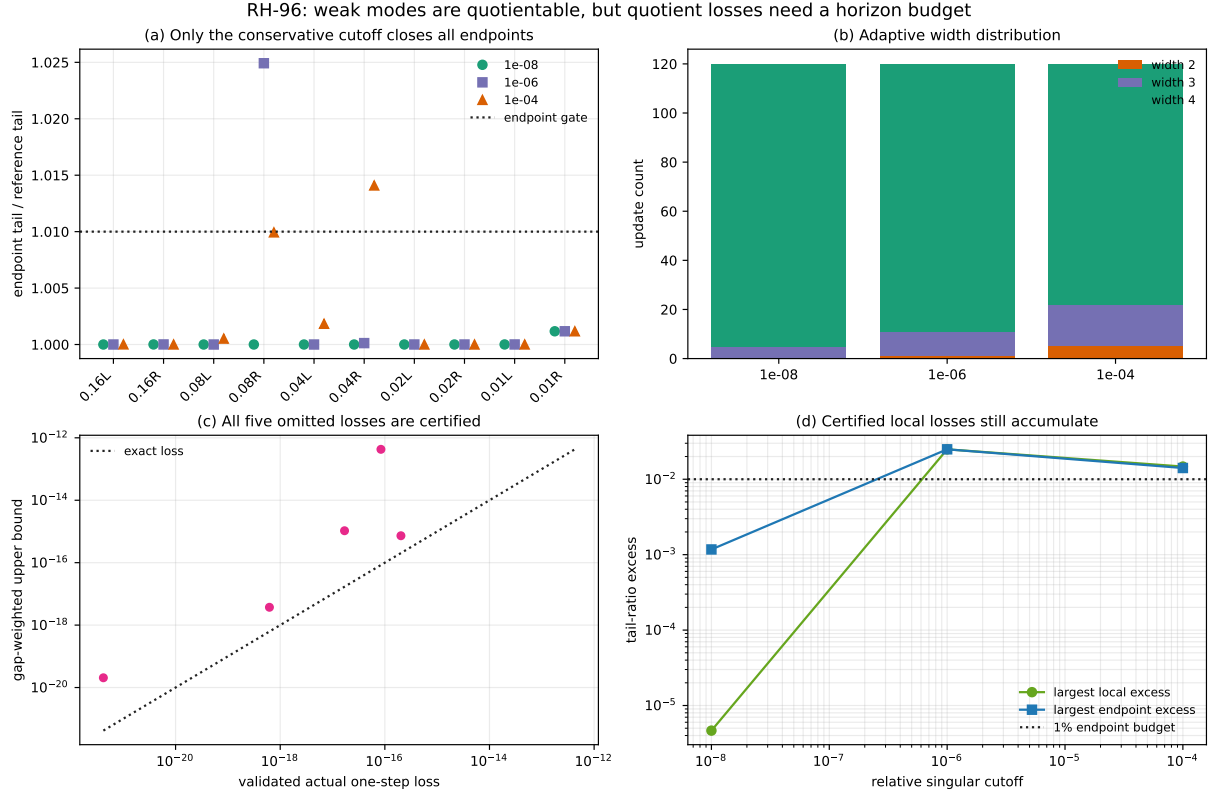


Figure 1: Weak-mode quotient audit. The  $10^{-8}$  cutoff removes exactly five ill-conditioned modes and preserves every endpoint. More aggressive local quotients remain individually certified but fail after recursive accumulation.

## 5 Toward a horizon quotient budget

Let  $E_t$  denote the tail excess of an adaptive packet relative to a reference full-width chain. The new theorem supplies a local injection  $\varepsilon_t$ , but a horizon theorem must also control sensitivity to an incoming packet error. The natural recurrence is

$$E_t \leq \rho_t E_{t-1} + \varepsilon_t, \quad (6)$$

where  $\rho_t$  measures predictor/refresh amplification. Iteration yields

$$E_N \leq \sum_{j=1}^N \varepsilon_j \prod_{\ell=j+1}^N \rho_\ell. \quad (7)$$

The RH-93 block budgets suggest that products of later  $\rho_\ell$  may be contractive even when individual factors exceed the desired target. A useful RH-97 result should therefore combine:

- the gap-weighted injections from Theorem 3.1;
- an a posteriori Lipschitz or projective sensitivity factor for one Ritz refresh;
- four-step products rather than rigid pointwise bounds;
- an endpoint budget stated relative to the ambient reference tail.

Such a theorem would explain why five tiny omissions survive but eleven larger ones do not.

## 6 Claim boundary

The universal omitted-block bound and the gap-weighted weak-mode tail-loss theorem are exact finite-dimensional statements. The  $10^{-8}$  adaptive chain is validated only on the ten frozen horizons. We do not prove:

1. a uniform retained-to-omitted gap or a universal relative cutoff;
2. a horizon composition inequality of the form (6);
3. a uniform adaptive-width or repeated-block contraction law;
4. an ambient-free continuum implementation;
5. Stage-A closure, a Hilbert–Polya operator, zeta-zero identification, or the Riemann Hypothesis.

## 7 Conclusion

The nearly null fourth projected-cross mode need not be reconstructed. A gap between the retained rank- $r$  Ritz cutoff and the omitted diagonal spectrum converts its coupling into the explicit loss bound  $\|C\|_F^2 / (\alpha - \beta)$ . At cutoff  $10^{-8}$ , this quotient removes all five ill-conditioned modes and preserves every source-to-endpoint gate.

The aggressive-cutoff experiments locate the next wall just as clearly: local certificates do not compose automatically. The route now needs a Duhamel-type horizon budget that transports each certified quotient loss through subsequent reduced Ritz updates.

## References

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